

BLE2LoRa: Cross-Technology Communication from Bluetooth to LoRa via Chirp Emulation

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Abstract—Wireless Personal Area Network (WPAN) technologies (e.g., Bluetooth, ZigBee) have been widely used in our daily life. However, due to their short transmission distances, the delivery of urgent messages (e.g., emergency alarms) over long distance, suffers from long delay, since the messages have to be transported via multi-hop way. Recent studies show that adding low-power wide-area network (LPWAN) radios such as LoRa onto WPAN devices (e.g., Bluetooth) effectively overcomes the limitation. However, the introduction of heterogeneous communication inevitably incurs extra hardware cost, deployment inconvenience and traffic overhead from gateways. In this paper, we present BLE2LoRa, a novel Bluetooth Low Energy (BLE) to LoRaWAN cross-technology communication (CTC) approach, which leverages the frequency shifting ability of BLE device to emulate LoRa's chirp signal. Such emulation is feasible because a chirp is a signal whose frequency increases or decreases over time, while a BLE device can also construct specific signals with ladder-shaped frequencies which is similar to LoRa chirp by carefully selecting BLE payload bits. Therefore, without any hardware modification at both sender and receiver side, a LoRa device can demodulate BLE frames. Moreover, leveraging the high sensitivity of LoRa base station, a long distant CTC can be achieved. We build our BLE2LoRa prototype on USRP B210 (with LoRaWAN PHY) and commodity BLE chips (CC1200). Our evaluation reveals that BLE2LoRa can achieve 4.06kbps throughput from BLE to LoRa with more than 80% frame reception rate, leading to over 600 meters communication distance, which is over 20x range extension over native Bluetooth.

Index Terms—cross-technology communication, long range, signal emulation, LoRa, Bluetooth

I. INTRODUCTION

Internet of Things (IoT) is revolutionizing the way we live and work [1], and has been widely deployed in many applications including smart city, intelligent industry and precision agriculture. To connect IoT, many wireless technologies have been applied such as WiFi, Bluetooth and LTE. Among them, WPAN technologies (e.g., BLE, ZigBee) gain their popularity due to their low-power and low-cost features. It is reported [2] that more than 4 billions of BLE devices have been used in our daily life including smart phones and smart watches, and more than 130 millions of ZigBee devices are deployed in smart homes, environment monitoring systems, etc.

WPAN technologies are designed for relatively short distance and high throughput transmissions, and their commu-

nication distances are limited to tens of meters. However, in many applications, IoT devices are sparsely deployed in a wide area. To provide long range connectivity, multi-hop mesh networking is proposed by researchers. The multi-hop transportation of messages, however, suffers from long delay, especially undesired for those messages carrying urgent information such as emergency alarms. Moreover, multi-hop transmission increases energy consumption on relay nodes and reduces the end-to-end reliability. For example, UC Berkeley [3] constructs a 48-hop wireless network to cover the Golden Gate Bridge which spans 4,200 feet, achieving only an 800bps low throughput to guarantee end-to-end reliability.

To overcome this problem, recent studies [4] have shown that adding low power wide-area network (LPWAN) radios such as LoRa [5] onto the WPAN (e.g., BLE) effectively overcomes its limitations, as the urgent messages can be transmitted directly to the base station within one hop. However, the heterogeneous network has to introduce gateways, which inevitably incur extra hardware cost, deployment inconvenience and traffic overhead.

Recent advances on Cross-Technology Communication (CTC) have built direct communications across heterogeneous technologies, which removes the need for gateways and therefore avoid their drawbacks. As a promising CTC technique, signal emulation achieves high throughput and transparent direct communications across several heterogeneous technologies, e.g., WiFi to ZigBee [6], LTE to Bluetooth [7], etc, by using a high-speed radio to fake the signals of a low-speed radio through manipulating its payload bits. However, the existing solutions are not directly applicable to emulate LoRa signals because of the unique characteristics of LoRa chirp. More specifically, the frequency within one LoRa chirp varies over time linearly. While for other widely-used modulation schemes (e.g., QAM, FSK, OQPSK), the frequency remains constant within a symbol/bit, although different frequencies can be used to modulate different symbols/bits. This technical gap undoubtedly brings about barriers and difficulties in emulating LoRa chirp from other technologies.

To deal with this challenge, in this paper, we take the first step towards emulating a frequency varying chirp. The key innovation is that we leverage the frequency shifting feature of BLE to construct specific signals with ladder-shaped spectrum to resemble the spectrum of LoRa chirp.

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More specifically, BLE devices utilize Frequency-Shift Keying (FSK) to modulate the payload bits into sine wave signals with different frequency shifts. Thus, we can emulate a chirp by manipulating the payload bits in a BLE frame to generate signals with incremental or decremental frequencies. At the LoRa receiver, when multiplying a down-chirp with the incoming signals and performing FFT, the receiver can identify the *chirp-like* BLE signals with high sensitivity and then demodulate it. Notice that the chirp spread spectrum (CSS) modulation of LoRaWAN leads to a high receiver sensitivity to -134 dBm [5], thus we can achieve a link budget closed to 134dBm from BLE sender (0dBm TX power) to LoRaWAN receiver via such chirp emulation, leading to much longer communication distance (theoretically over 40 kilometers if perfectly emulated) than native BLE communications.

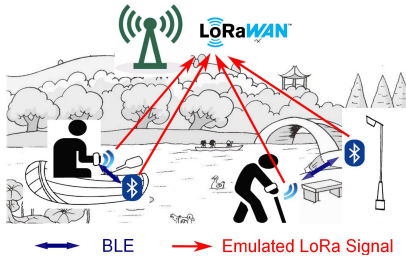


Fig. 1. Scenarios of BLE to LoRa CTC applications

With this long range CTC, BLE devices can report their key messages to a far away base station within only one hop. By removing the heavy cost of multi-hop relays, the transmission delay can be significantly reduced and the end-to-end reliability can be improved. This technology, potentially, could enable a set of new IoT applications including, but not limited to, health care, maintenance notification, and disaster communications. For example, as illustrated as Fig. 1, a wearable BLE device can send an emergency message [8] from an old man to a remote base station for health care when he is far away from home. A street lamp equipped with BLE radio [9] can report its fault diagnosis and call for timely maintenance. A boat with BLE radio can report to a base station even it flows far away along a river, etc. In a word, the direct communication between LoRa and BLE can take advantages of both the ubiquity of BLE devices and long range transmission ability of LoRa.

The major contributions of BLE2LoRa are as follows:

- We present BLE2LoRa, a novel long range CTC technique from BLE to LoRa. The unique design of BLE2LoRa lies on *chirp emulation*, that is, constructing specific BLE signals with ladder-shaped frequencies by BLE payload selection, to emulate desired LoRa chirps with linearly changed frequencies. BLE2LoRa is a transparent design that requires no hardware modification at both sender side and receiver side.
- We address several important challenges, namely (i) how to synthesize a valid LoRa frame from BLE signals under the constraints of LoRaWAN physical layers, (ii)

how to achieve symbol-level synchronization using the sync word design, and (iii) how to decode the imperfect emulated chirps with reliable decoder. These techniques provide general guidance for future CTC designs from WPAN (e.g., ZigBee) to LPWAN (e.g., NB-IoT).

- We implement BLE2LoRa on USRP-B210 (with LoRaWAN PHY) and commodity BLE chip (CC1120). Our extensive evaluations demonstrate that BLE2LoRa achieves direct communication from a BLE sender to a LoRaWAN receiver at 4.06kbps with more than 80% frame reception rate. The transmission distance is more than 600 meters, dramatically outperforming the native BLE communication.

II. OVERVIEW

A. BLE2LoRa In a Nutshell

Fig. 2 illustrates how BLE2LoRa works: a BLE device transmits a frame with chosen payload that can be recognized by a LoRa receiver as a valid LoRa frame. Specifically, BLE2LoRa constructs the payload of a BLE frame so that a sequence of emulated LoRa chirps is synthesized by the modulation of BLE devices (i.e., FSK). When such a BLE frame is transmitted into the air, the original BLE preamble, BLE header, and BLE trailer are ignored by LoRa receivers as noise, while the chosen payload emulating the LoRa chirps will successfully pass the LoRa preamble detection, demodulated and decoded by LoRa receiver, and then assembled into a legitimate LoRa frame.

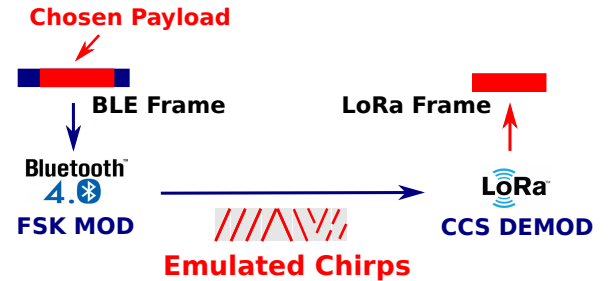


Fig. 2. Overview of BLE2LoRa

B. Preliminary: CSS modulation and demodulation

The LoRa physical layer modulates the information in the form of chirp pulses, where a chirp is a kind of signal whose frequency varies linearly in time over the available bandwidth. Fig. 3 illustrates two type of chirps, i.e., the up-chirp and the down-chirp, in the time domain, and their corresponding spectrogram. Up-chirp is a signal with increasing frequency and down-chirp is the one with decreasing frequency. They are two fundamental chirps on which the CSS modulation and demodulation are designed.

CSS modulation leverages the baseline up-chirp to depict symbol '0'. And other symbols of CSS are distinguished by initiating the up-chirp at different frequencies. For instance, the up-chirp beginning at the half of the bandwidth can be leveraged to represent symbol '1' if the spreading factor

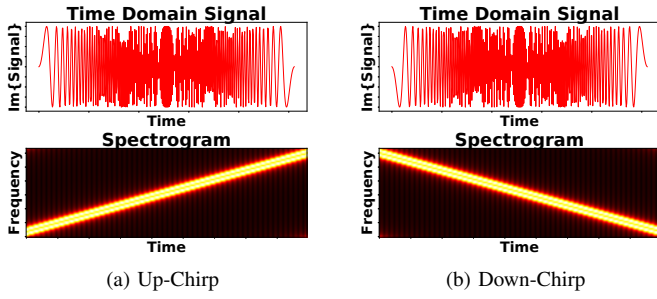


Fig. 3. The Up-Chirp and Down-Chirp of CSS

(i.e., SF) is set to 1. Fig 4a shows the spectrograms of the two chirps corresponding to symbol ‘0’ and symbol ‘1’. The observation leveraged by CSS demodulation is that the up-chirps with different initial frequencies will lead to the cross-correlation peaks at different locations, illustrated as Fig. 4b, when cross-correlated with the baseline up-chirp signal. Thus, by obtaining the location of the cross-correlation peak, the CSS demodulator demodulates different chirps.

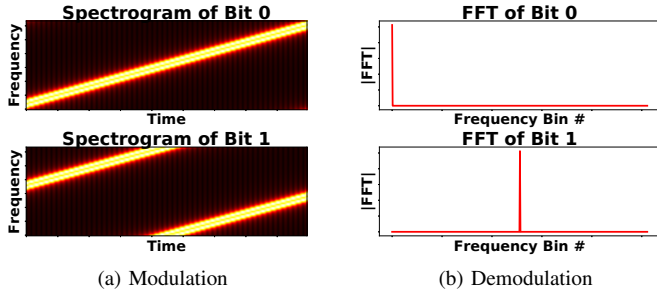


Fig. 4. Modulation and Demodulation of CSS

We note that, in signal processing, the cross-correlation of two signals is the sliding inner-product of one signal and the complex conjugate of another signal. For the baseline up-chirp shown in Fig. 3a, its complex conjugate is the down-chirp signal shown in Fig. 3b. They are given by:

$$Chirp_{up}(t) = e^{2\pi j(\lambda t^2 + \phi t)}, \quad Chirp_{down}(t) = e^{-2\pi j(\lambda t^2 + \phi t)}. \quad (1)$$

where λ and ϕ are two parameters decided by the spreading factor (SF) and the initial frequency. Meanwhile, in signal processing, the multiplication in the time domain is the convolution in the frequency domain. Thus, the LoRa receiver demodulates the up-chirps by first multiplying them with a baseline down-chirp in the time domain and then performing Fast Fourier Transform (FFT). Due to conjugation with each other, the down-chirp and the up-chirp have a very strong correlation, so the operation of above FFT results in a peak at an FFT frequency bin. The up-chirps with different initial frequencies, which are also the time shifted up-chirp with different time delays, leads to occurrence of a peak at different frequency bins. For example, as illustrated ins Fig. 4b, the demodulation of the chirp signal corresponding to symbol ‘0’ results in a peak at the 0th bin, while the peak for symbol ‘1’ is located at the middle of FFT bins. CSS demodulation, in a nutshell, localizes the single peak on the result of the FFT.

C. Chirp Emulation by BLE GFSK

To explain how to emulate chirps with GFSK, first we show how a BLE transmitter produces single tone sine-waves. With GFSK modulation, the data bits in a BLE frame are modulated into the physical signals as following:

$$I(t) = \cos(\phi(t)), \quad Q(t) = \sin(\phi(t)) \quad (2)$$

$$\phi(t) = \pi h \int_0^t \sum_{i=0}^{\infty} d_i g(\tau - iT_b) d\tau,$$

where $I(t)$ and $Q(t)$ are the in-phase and quadrature branches of the BLE baseband signals. d_i denotes the i_{th} data bit, T_b represents the duration of a BLE symbol, and g is the Gaussian function for pulse shaping. The modulation index, h , which controls the shape of the modulated signals, is 0.5 in BLE.

Using GFSK modulation, a BLE device is able to generate some specific time domain waveforms if the data bits are chosen elaborately. For example, if the data bits of a BLE frame are all 1, the phase of the signal $\phi(t)$ will be $\pi h t / T_b$, then the time domain signal transmitted from the BLE modulator is given by $I(t) = \cos(\frac{\pi}{2T_b} t)$, $Q(t) = \sin(\frac{\pi}{2T_b} t)$, i.e., a sine wave with single tone. Therefore, when a bit stream with all 1 is generated, a sine wave with a frequency shift as $\frac{\pi}{2T_b}$ (i.e., the right tone in Fig. 5) is produced. The same way, while for a bit stream with all 0, a sine wave with a frequency shift as $-\frac{\pi}{2T_b}$ (i.e., the left tone in Fig. 5) is produced.

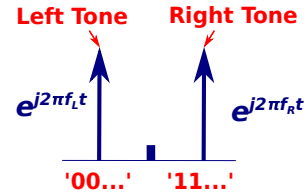


Fig. 5. Specific BLE waveforms with two tones

Leveraging this frequency shifting feature, we can generate the waveform with some expected frequency patterns by elaborately choosing the bits of the BLE frame. For instance, when BLE bits with increasing frequencies are sent, a sequence of signals resembling the up-chirp waveform will be generated. Fig. 6 illustrates an emulated up-chirp from BLE modulated by 4-FSK¹. A frame with payload “00 01 10 11” is able to produce a section of signal with upward staircase spectrogram (Fig. 6b), as the bits of “00” in 4-FSK modulation produce a sine wave with lowest frequency, and the bits of “01” produce the sine waves whose frequency is higher, and so on. Such staircase spectrogram can emulate the up-chirp signal (Fig. 3a) to a certain extent because their spectrum are similar.

The result in Fig. 7 proves that such emulation is feasible. We note that not only the emulated signal looks very like the up-chirp signal in time domain (Fig. 7a), but also when we demodulate the emulated signal by multiplying it with a down-chirp signal and then performing FFT, the result (Fig. 7b)

¹we note that some BLE commodity chips (e.g., CC1200, CC1120, etc.,) utilize 4-FSK as an optional modulation scheme.

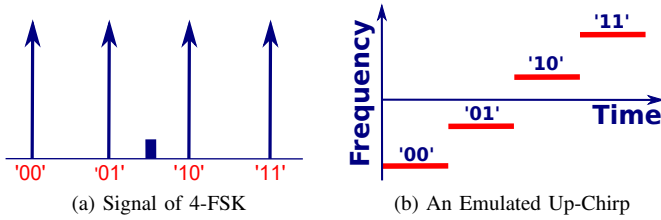


Fig. 6. Chirp Emulation using FSK Modulation

shows that the emulated signal leads to the same peak as the native up-chirp, thus enabling successful demodulation at LoRa receiver. Though such emulation may bring some distortions as shown in Fig. 7b, the advanced designs described in Section III enables the demodulator and decoder of BLE2LoRa to eliminate the errors caused by imperfect emulation.

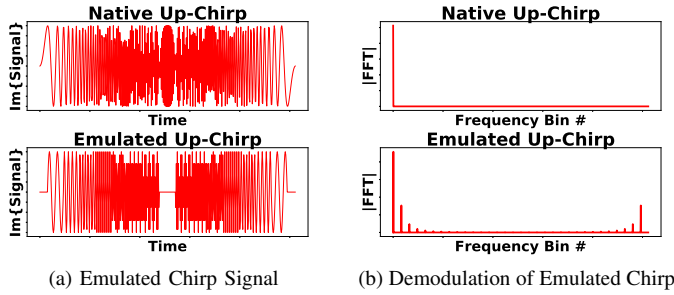


Fig. 7. Proof of Concept Example for Chirp Emulation

III. CHALLENGES AND SOLUTIONS

To reliably emulate a LoRa frame on real-world BLE devices, there are still a few challenges to deal with. More specifically, under the constraints of LoRa physical layers, e.g., the bandwidths, the symbol rates, how to choose appropriate PHY parameters to synthesize a frame is challenging. Moreover, the imperfection of the emulated chirps leads to bad performance in both synchronization and demodulation. Therefore, we need to deal with two other challenges, i.e., how to synchronize the emulated LoRa chirps accurately and how to decode the imperfect emulated chirps reliably. In the next subsections, we explain these challenges and their corresponding solutions.

A. Challenge I: Synthesizing BLE Frame for Emulation

To achieve the physical-layer CTC on real LoRa and BLE devices using chirp emulation, it is necessary to satisfy the physical layer constraints on both BLE sender and LoRa receiver. More specifically, when an emulated chirp is transmitted by BLE, both its frequency range and its time duration has to match with the target LoRa chirp.

In real scenarios, LoRa devices operate at configurable data rates, by adjusting two basic parameters: Spreading Factor (SF) and Bandwidth (BW). Spreading factor determines how many bits can be transmitted by one LoRa chirp. For example, when a LoRa device adopts a spreading factor of 6 (i.e., SF = 6), it will choose one of the 2^6 optional starting frequencies (i.e., 2^6 states) to modulate an up-chirp, that is, 6 bits is

transmitted per chirp. The bandwidth, on the other hand, determines the range and the slope that the frequency of the chirp varies.

As we know, LoRa standard allows a spreading factor from 6 to 12. For better illustration, here we choose the spreading factor of 6 in chirp emulation due to the fact that the fewer optional starting frequencies are, the larger emulation distortion is tolerable². LoRa also operates with eight chirp bandwidths between 7.8 kHz and 500 kHz. We note that the frequency deviation of FSK adopted by commodity BLE device is ± 250 KHz, so the chirp with bandwidth of 500 KHz is adopted since 500 KHz is an integral multiple of 250 KHz. In LoRa, a chirp has 2^{SF} samples, and the sampling rate is equal to its bandwidth. Therefore, if the above two fundamental parameters, i.e., SF = 6 and BW = 500 kHz, is adopted, the duration of an emulated chirp is given by:

$$T_{chirp} = \frac{2^{SF}}{BW} = \frac{64}{500KHz} = 128\mu s \quad (3)$$

BLE device normally adopts the symbol rate of 1 Msym/s. Therefore we need to utilize 128 BLE symbols to emulate a LoRa chirp. For the 4-FSK modulation (Fig. 6), 2 bits are modulated into a BLE symbol, and thus 256 BLE bits are utilized to emulate a LoRa chirp. For example, if we want to emulate the baseline up-chirp using the staircase spectrogram shown as Fig. 6, each of four lines is emulated by 64 BLE bits (32 symbols). The first line is emulated by 32 repeated BLE symbols as '00', and the second line is emulated by 32 repeated BLE symbols as '01', etc. In the same way, if we want to emulate the up-chirp with the first starting frequency among 2^6 options, we can cyclically left-shift the above BLE bit sequence by 4 bits, because the cyclic left-shifting of the baseline up-chirp by $\frac{128\mu s}{64} = 2\mu s$ is equal to cyclic left-shifting of the BLE sequence by 4 bits.

To implement our system on a real platform, there is still a practical issue to be addressed. Due to security issue, the bits transmitted by a BLE device are not the original payload bits. Before modulation, the payload bits will pass through a scrambler, also known as data whitening module. Upon receiving the data bits, the receiver needs to reversely convert the raw data bits back into the payload bits through the descrambler. So it is necessary to consider the data whitening when selecting the payload bits at BLE transmitter.

In practice, BLE uses a linear-feedback shift register (LFSR) as a scrambler, which scrambles the source bits by XORing them with a pseudo random noise (PN) sequence. The scrambling seed is its operating channel. Therefore, we can reversely construct the BLE payload bits by inputting the desired raw BLE bits into the BLE scrambler and then output the BLE payload bits.

B. Challenge II: Chirp-Level Synchronization

To emulate a valid LoRa frame, the BLE sender has to construct a specific frame structure as defined by LoRa. Fig. 8

²we note that when other spreading factors are adopted, the emulated chirp can also be decoded using the reliable decoder discussed later.

illustrates the structure of a LoRa frame. As shown, the LoRa frame begins with a preamble consisting of several baseline up-chirps, then a sync word of two baseline up-chirps, followed by two and a quarter down-chirps for chirp-level synchronization. Then, it is followed by the payload chirps and the CRC chirps at the end of the frame. When receiving a frame, the LoRa receiver first detects the repeated baseline up-chirps (i.e., the preamble). After that, the LoRa receiver localizes the start point of the down-chirp for accurate chirp synchronization. Based on the start point of the down-chirp, the LoRa receiver divides the received signal samples into segments as individual chirps, and then demodulates each of the up-chirps with different starting frequencies into LoRa symbols one by one in the payload. Finally, the LoRa receiver decodes them into a data block for CRC checking.

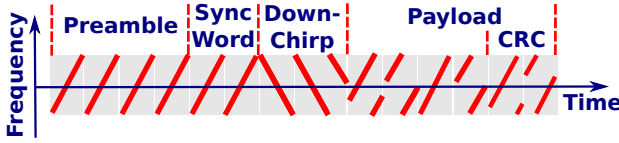


Fig. 8. The Structure of LoRa Frame

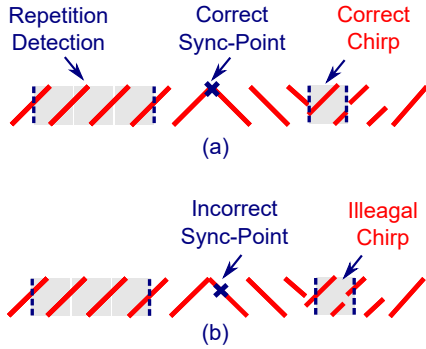


Fig. 9. Why Synchronization is Important

As described above, the performance of LoRa demodulation depends heavily on the accuracy of the chirp synchronization. For example, as shown in Fig. 9(a), if the starting point of the down-chirp is localized at the right place, then the following signal samples can be correctly divided into individual chirps for demodulation. On the other hand, as shown in Fig. 9(b), if the starting point of the down-chirp is incorrectly localized, then the boundaries of all following up-chirps will be wrong, leading to erroneous demodulation.

Fig. 10 illustrates the procedure of LoRa synchronization. To localize the position of down-chirp, the LoRa receiver continuously correlates a part of the received signal with the down-chirp template in a sliding window (the shadow box in Fig. 10). The sliding window of the received signal moves forward each time with a small time slot. For a native LoRa frame, the starting point of the down-chirps (i.e., the Sync-Point t_4 in Fig. 10) can be easily found, because at this moment, the signal can best match with the down-chirp template, resulting in a sharp correlation peak at t_4 . However, for the emulated LoRa frame (Fig. 11), the emulated down-

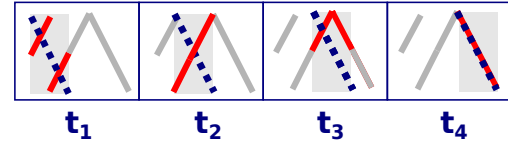


Fig. 10. How LoRa Synchronization Works.

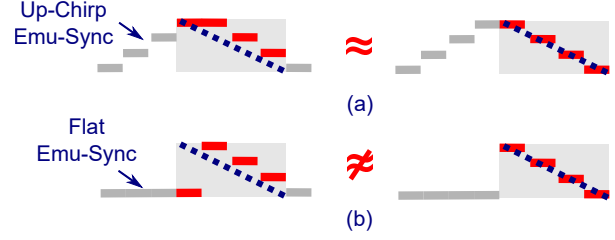


Fig. 11. Comparison of Two Sync Words.

chirp is no longer a perfect descending line but a staircase signal in frequency domain, reducing the synchronization accuracy as a down-chirp.

The reason is further explained as follows. As illustrated in Fig. 11(a), when an emulated up-chirp is utilized as the sync word, the last quarter of this up-chirp will be connected with the beginning quarter of the emulated down-chirp, thus forming a flat signal, i.e., a line in frequency domain. In this case, if we move the sliding window from the start point of this line to the end of it, and perform correlation with a standard down-chirp, the correlation results will be almost the same. Therefore it is hard to localize the exact starting point of the emulated down-chirp (i.e., the middle point of the line) for accurate synchronization.

To solve this problem, we leverage a flat signal (i.e., a line in frequency domain) instead of an emulated up-chirp as sync word, as shown in Fig. 11(b). With this sync word, the starting point of the emulated down-chirp can be easily identified since the signal changes abruptly at this point. After correlation and performing FFT, the results will also suddenly change at this point, leading to a much better performance for synchronization. The experimental results of Fig. 12 show that the flat sync word leads to the same sync position (i.e., the time slot when the value of $\text{argmax}(\text{FFT})$ start to change slowly) as that of native chirps. In contrast, when the emulated up-chirp is utilized as a sync word, the bins of maximum FFT always change slowly when the position of time slots varies, make it vulnerable to synchronization errors.

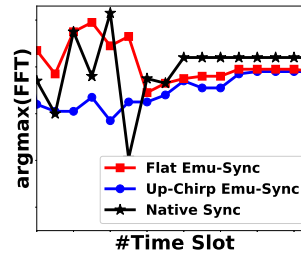


Fig. 12. Effect of Two Emulated Sync Words

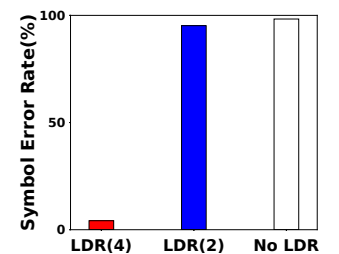


Fig. 13. Performance of Cluster-Based Decoder

C. Challenge III: Decoding Imperfect Emulated Chirps

Though the correct boundaries of the emulated chirps are found by redesigning the sync word, the chirp emulation may still lead to erroneous demodulation because of the emulation distortions shown as in Fig. 7b. It is also shown from Fig. 7b that the FFT values corresponding to the correlation are clustered. Based on such observation, we design a cluster-based decoder to deal with the imperfect chirp emulation at the cost of decreased data rate. More specifically, considering the maximum FFT magnitude, several FFT bins (e.g., four FFT bins) around the supposed bin are decoded as the same bits.

We note that LoRa provides a mode of Low Data Rate (LDR) optimized for improving the reliability, and some commodity LoRa chips (e.g., Semtech's transceivers [10]) implement this mode. LDR is actually a cluster based decoder. Fig. 13 illustrates the demodulation performance of LDR with different parameters, i.e., LDR = 0 (no LDR), 2 and 4. Specifically, for the BLE2LoRa without cluster based decoder (no LDR), and LDR with cluster size as 2, the symbol error rate (SER) are both too high ($>90\%$) to achieve a feasible communication from BLE to LoRa. However, when the size of the cluster in LDR is set to 4, the performance becomes good (i.e., $SER < 4\%$) instantly.

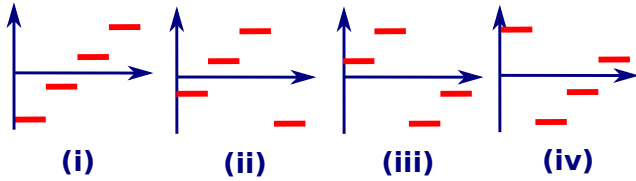


Fig. 14. Four Most Significant Shifts for the Emulated Chirps

The reason for such a leap in performance is that the parameter of LDR (i.e., LDR(4)) matches the chirp emulation utilized in BLE2LoRa. More specifically, there are $2^6 = 64$ up-chirps in total when the spreading factor is 6. Meanwhile, the LDR optimization with parameter of 4 leads to a clustering on these up-chirps where each cluster has a size of $2^4 = 16$. Therefore, the emulated chirps, using such emulating parameters, are divided into $\frac{64}{16} = 4$ clusters. Illustrated as Fig. 14, these four clusters exactly match the four most significant time-domain shifts when the 4-FSK modulation emulates a spectrogram comprising 4 stairs. That is, BLE2LoRa leverages four signals shown in Fig. 14 to convey the messages in a reliable way. Now, BLE2LoRa achieves a data rate of $1/T_{chirp} \times (Bits/Chirp) = \frac{1}{128\mu s} \times 2 = 15.6$ Kbps.

IV. DISCUSSION

A. Bidirectional Links

For the real implementation of our system, a downlink communication (i.e., from LoRa base station to BLE devices) is typically required. Especially, we may need an ACK as a feedback to ensure reliable transmission. This downlink CTC could be easier to be achieved than the uplink CTC, because a base station generally has much richer resources (e.g., wider

bandwidth) and possesses much stronger capabilities (e.g., higher transmission power) to deliver side channel messages to end devices. For example, LoRaBee [11] achieves LoRa to ZigBee CTC by putting specific bytes in the payload, which are selected such that the corresponding LoRa chirps can be recognized by the ZigBee devices through sampling the received signal strength (RSS). Since many commodity BLE radios (e.g., CC2540) have the same ability to sample RSS, other packet-level CTC schemes (e.g., [12], [13]) can also be adopted to achieve LoRa to BLE CTC.

B. Concurrent Transmissions

Similar to LoRaWAN, BLE2LoRa adopts ALOHA, a random channel access MAC layer protocol without carrier sensing. With ALOHA, BLE devices transmit at any time it has data to send. This may lead to collisions among different transmissions. To alleviate the problem, we make BLE devices randomly select a channel frequency from the list of available channels in the native LoRaWAN, then randomly select a SF between 7 and 12. Since those SFs are orthogonal at the same channel, simultaneous transmissions only cause a collision when they both select the same SF and the same channel, respectively. A typical LoRa base station is equipped with 8 RF radios operating at different frequency channels, which further enables BLE2LoRa to support concurrent transmissions from multiple BLE devices. If a collision still happens, the BLE devices will hop to other channels and retransmit the packet.

V. PERFORMANCE EVALUATION

As shown in Fig. 15, we implement BLE2LoRa on a USRP-B210 platform with LoRa PHY [14] and commodity BLE chips (CC1200). We compare BLE2LoRa with two baselines: a BLE communication on commodity chips (i.e., BLE chip CC2540) and a LoRa communication on USRP platforms. To evaluate the performance of BLE2LoRa, the transmission power of BLE2LoRa and two baselines are all fixed as 0 dBm, a typical power configuration adopted in low power wireless devices.

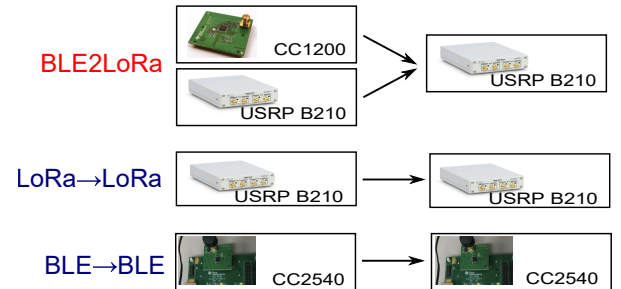


Fig. 15. Experiment Setting for BLE2LoRa

In the experiments, each emulated frame consists of eight up-chirps as a preamble, 2 up-chirps as a sync word and $2\frac{1}{4}$ down-chirps for synchronization, and a default frame payload of 30 bytes. To ensure statistical validity, we obtain the average result of 10 experiments, each of which sends 1,000 BLE2LoRa/BLE/LoRa frames under a wide range of settings including indoor/outdoor, stationary and mobile scenarios.

A. Long Range Performance

The evaluation of BLE2LoRa mainly focuses on the link-layer statistics (i.e., frame reception ratio (FRR)) in long range communication. We conduct comprehensive experiments in several typical sites, including (i) a LOS site on an urban road (Fig. 16), and (ii) a NLOS site in a campus (Fig. 19).



Fig. 16. Road Site (Map).

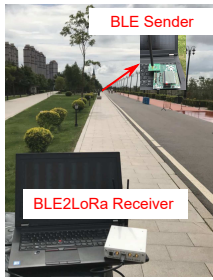


Fig. 17. Road Site (Picture).

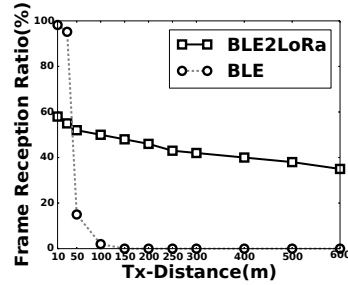


Fig. 18. Frame Reception Ratio with Tx Distances (Urban Road).

We first evaluate the long range performance of BLE2LoRa at a site of a road (Fig. 17) in a city. We set the transmission power of BLE2LoRa and BLE at 0dBm and only change the transmission distances. Fig. 18 shows the FRR of BLE2LoRa and BLE under varying distances. From the experimental results, we can see that BLE2LoRa achieves a good emulation (i.e., $FRR > 30\%$) even the transmission distance is above 600 meters, close to the emulation performance when the sender and receiver are near ($< 100m$). In contrast, the FRR of BLE drops to about 10% when the communication distance is above 50 meters, though its FRR is more than 95% for the distance less than 30 meters. For short distance, BLE2LoRa is not as good as BLE because of its considerable emulation distortion. However, when the transmission distance is large, BLE2LoRa greatly outperforms BLE due to the effect of the higher receiver sensitivity from the emulation of LoRa chirp.

We also evaluate the long range performance of BLE2LoRa in a non-line-of-sight (NLOS) scenario, i.e., a campus site shown in Fig. 19. We place the BLE2LoRa receiver (i.e., the LoRa base station) on a windowsill of a laboratory at the 8th floor of a building (the height is about 30 meters), and place the BLE senders at the location A, B, C, D, E, F illustrated as Fig. 19 respectively. We note that the distances between the receiver and the transmitters are 30 meters (A), 130 meters (B), 220 meters (C), 300 meters (D), 400 meters (E) and 420 meters (F) respectively, and the testing locations D, E, and F are in NLOS conditions. The experimental result (Fig. 20) illustrates that the FRRs of BLE2LoRa either at

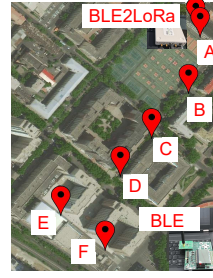


Fig. 19. Campus Site (map).

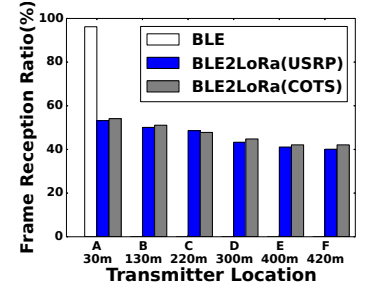


Fig. 20. Frame Reception Ratio with Tx Distances (Campus).

USRP platform or at the COTS devices, are more than 40% even the transmission distance is above 400 meters and the communication is under the NLOS condition, indicating that BLE2LoRa can effectively support long range communications for complex environments.

B. Experimental Insights

To understand why BLE2LoRa works, we conduct another set of experiments to obtain the physical-level metric, i.e., the receiver sensitivity, of BLE2LoRa. To eliminate the influence of the irrelevant factors including platform, radio, antenna, etc., we measure the relative sensitivity of BLE2LoRa in comparison to LoRa, where BLE2LoRa and LoRa work at the identical settings except the transmission power. We set the transmission power of BLE2LoRa as 0dBm and the distance as 400m where FRR is shown to be 40%, and then change the transmission power of LoRa.

Fig. 21 and Fig. 22 show the FRR of LoRa when only the difference of transmission power between BLE2LoRa and LoRa (i.e., $\Delta Tx\text{-Power} = Tx\text{-Power(LoRa)} - Tx\text{-Power(BLE2LoRa)}$), varies from -4dBm to -15dBm. Fig. 21 shows the result at a site under line-of-sight, and Fig. 22 provides the result at a non-line-of-sight site. Experimental results show that the receiver sensitivity of BLE2LoRa is about 12dBm (LOS) and 8dBm (NLOS) worse than those of native LoRa. Due to the intrinsic errors caused from the signal emulation, BLE2LoRa with FRR of 40% is compared to native LoRa with FRR of 80%.

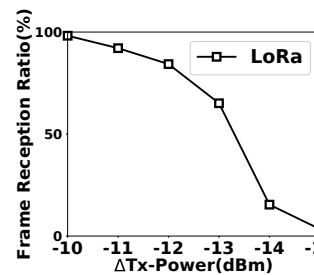


Fig. 21. Changes in LoRa's FRR with $\Delta Tx\text{-Powers}$ (LOS).

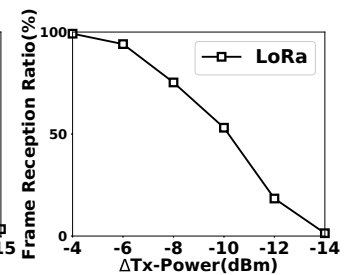


Fig. 22. Changes in LoRa's FRR with $\Delta Tx\text{-Powers}$ (NLOS).

The results of Fig. 21 and Fig. 22 prove that BLE2LoRa has a moderate deterioration (about 10 dBm) in receiver sensitivity in comparison to those of LoRa, and such deterioration is

stable in different experimental scenarios. We note that LoRa typically has the receiver sensitivity of -134 dBm [5]. Thus the receiver sensitivity of BLE2LoRa can be as low as -124 dBm, while the receiver sensitivity of BLE only reaches -93 dBm, enabling BLE2LoRa to achieve 40x range extension over BLE in theory (Eq. 1) and over 20x at experimental sites.

C. Reliable BLE2LoRa with Repeated Transmission

Native BLE2LoRa achieves about 40% FRR when transmission range is more than 400 meters. However, we find from the experimental results that almost all the preambles of those frame (> 98%) are detected successfully. Therefore, we only need to rectify the possible emulation errors in payload symbols. We can utilize coding schemes (e.g., Hamming Coding, RS coding) or repeated transmissions to improve the reliability. Retransmission is a simple and transparent design since it does not need any software modification at the receiver. Fig. 23 demonstrates how the reliability of BLE2LoRa can be improved when the number of repeated transmissions (#trans increases. The results show that BLE2LoRa achieves a reliable communication (>80%) if we repeat the transmission 4 times, with a more than 4kbps throughput (Fig. 24) which is comparable to other techniques for LPWAN applications [15]³.

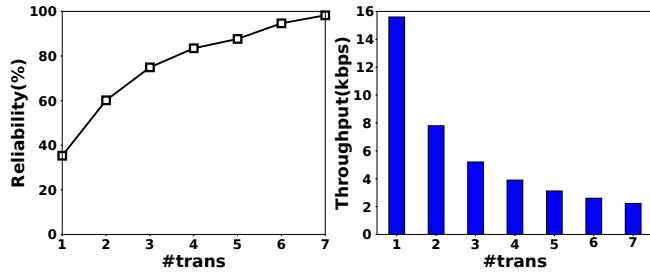


Fig. 23. Reliability of BLE2LoRa with Repeated Transmissions.

Fig. 24. Throughput of BLE2LoRa with Repeated Transmissions.

D. Flexibility on Commodity Devices

We also evaluate the implementation flexibility of BLE2LoRa on commodity BLE devices, i.e., CC1200. We note that native LoRa provides multiple spreading factors (SFs) and bandwidths (BW) to adapt to the scenarios where variable transmission distances and data rates are required. Though the RF parameters of 4-FSK CC1200 can not be arbitrarily set, we can still achieve some flexibility for BLE2LoRa communication based on the limited settings provided by CC1200.

More specifically, when CC1200 is set the frequency deviation as 250kHz, the symbol rate as 250ksps, and the modulation as 4-FSK, the emulated LoRa bandwidth will be 500kHz and each LoRa symbol (SF = 6) lasts $128\mu s$ ($\frac{2^6}{0.5}\mu s$). To emulate a LoRa chirp, we need 64 BLE bits

³With feedback messages (e.g., ACK or NACK) discussed in Sec. IV, the number of repeated transmissions can be significantly reduced. Also, coding schemes can significantly improve the efficiency for reliable transmission, which will be our future work.

($128\mu s \times 250ksps \times 2bits/symbol$). We, thus, send the sequence of hexadecimal digits “55 55 00 00 AA AA FF FF” (symbol encoding shown in Fig. 6a) using the CC1200, the LoRa receiver can obtain an approximate up-chirp with SF of 6 and BW of 500kHz. Similarly, if we set the frequency deviation of 125kHz (62.5kHz), the symbol rate of 125ksps (62.5ksps) for CC1200, we can emulate a LoRa up-chirp with SF = 6 and BW = 250kHz (125kHz). Fig. 25 shows that the BLE2LoRa on CC1200 can provide the flexibility like native LoRa [5].

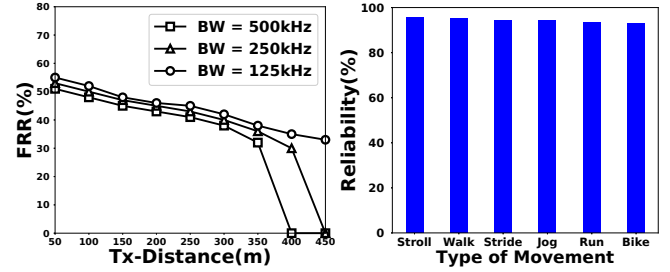


Fig. 25. FRR of BLE2LoRa (COTS) using Different Bandwidth.

E. Performance under Mobility

We also evaluate BLE2LoRa with a mobile BLE sender. We conduct the experiments at the campus site (Fig. 19) and set the transmission distance as 400 meters. Six levels of movements are evaluated in the experiments, namely, strolling, walking, striding, jogging, running and biking. In this set of experiments, each BLE2LoRa transmission is repeated 7 times. Fig. 26 shows that BLE2LoRa is able to work well under different levels of mobilities.

VI. RELATED WORKS

Nowadays, many WPAN technologies, such as Bluetooth, ZigBee, Z-wave [16] and WirelessHART [17], have been widely deployed in many IoT applications, including smart city, smart home, intelligent industry and environment monitoring. However, the performance of long range message delivery via multi-hop still remains unsatisfactory. Recently, Low-Power Wide-Area Network (LPWAN) emerges as a promising platform to offer wide-area connectivity for low power wireless devices. Several LPWAN protocols have emerged in the past few years, including those for the unlicensed band (LoRaWAN [5], SigFox [18]), and for the licensed band (LTE-M [19], NB-IoT [20]). Adding these LPWAN devices into the existing widely used WPANs is an appealing solution to improve the long range transmission performance of WPANs.

Recent advances on cross-technology communication (CTC) enables direct communication across heterogeneous wireless devices without gateways. Packet-level CTC (PKT-CTC) techniques leverage the packet-level information such as the received signal strength (RSS) [11], [12], [21], the transmission timings [22]–[24], the packet length [25], or the channel state indicator (CSI) [26], [27] to piggy-back side channel messages. Because each packet only modulates several bits, the throughputs of the packet-level CTCs are restricted to a few tens or hundreds of bps.

Among CTC techniques, signal emulation is an effective way to achieve high throughput and transparent communication across heterogeneous technologies, since they exploit physical layer signal to convey information directly. WEBee [6] is the first physical layer CTC (PHY-CTC) work which emulates ZigBee signal by manipulating WiFi OFDM symbols in payload. BlueBee [28] achieves PHY-CTC from Bluetooth to ZigBee by emulating phase shift sequence of ZigBee signal. WIDE [29] proposes the concept of digital emulation, which emulates ZigBee binary phase shift sequence instead of standard ZigBee waveform for ZigBee decoding. , to achieve a more reliable CTC. LTE2B [7] proposes time-domain signal emulation approach to achieve PHY-CTC from LTE to BLE. Depart from these works, BLE2LoRa emulates LoRa signals from BLE devices, which is unique in the way that the frequencies of LoRa chirp is varying linearly over time within a symbol. Symphony [30] is another work achieves long range CTC from low power devices (i.e., BLE and ZigBee) to LoRa base station. However, it has to modify the hardware of LoRa base station, thus is not fully compatible to the native LoRa standard.

VII. CONCLUSION

In this paper, we present BLE2LoRa, a novel CTC approach to enable the direct communication from BLE to LoRa. By choosing payload bits from a legitimate BLE frame, we generate BLE signals with ladder-shaped frequencies, which resemble LoRa chirps with linearly varying frequencies. As LoRa receiver can detect the emulated chirps with very high sensitivity, BLE2LoRa enables long range direct communication from BLE to LoRa. Experiment results show that BLE2LoRa achieves reliable data communication over 600m distance, extending the communication range more than 20x that of the native BLE.

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